TECHNICAL DATA SHEET
ProLine PPA-CF Filament

Product Overview

ProLine PPA-CF is a polyphthalamide-based material reinforced with 10% carbon fiber. It combines high thermal stability with low moisture uptake and reduced shrinkage. Parts printed from PPA-CF offer strong tensile and impact performance and can reach a heat deflection temperature of up to 220°C. Thanks to its high-temperature creep resistance, dimensional stability and electrical characteristics remain comparatively consistent across changes in humidity and temperature.

Key Features

- Rigid, dimensionally stable parts
- Strong, durable components
- Performs reliably at elevated temperatures
- Good creep resistance for long-term loads

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm³	1.21–1.23
MFR (280°C / 5 kg)	ISO 1133	g/10 min	35–40
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.1
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa	108–112
Elongation at break (X-Y)	ISO 527	%	3.5–5
Flexural modulus (X-Y)	ISO 527	MPa	7100–8000
Flexural strength (X-Y)	ISO 178	MPa	210–221
Flexural strength (Z)	ISO 178	MPa	70–75
Flexural modulus (X-Y)	ISO 178	MPa	6500–7000
Flexural modulus (Z)	ISO 178	MPa	2520–2620
Impact strength (X-Y)	ISO 180	kJ/m²	9–11
Notched impact strength (Z)	ISO 180	kJ/m²	2.5–3
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	220

Continuous Use Temperature	IEC 60216	°C	120

Electrical characteristics			
Surface resistance	IEC 60093	Ω	≤ 10^3

Recommended Printing Parameters

Parameter	Setting
Print Temperature	290-310°C
Bed Temperature	100–120°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	0–30%
Printing Environmental Temperature	Room temperature to ~60°C

Drying & Handling Notes

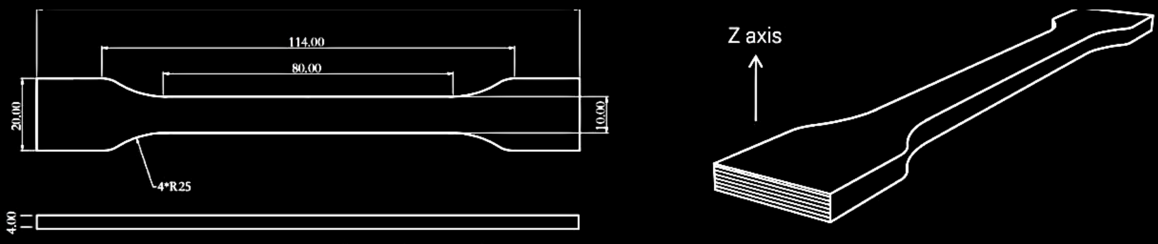
- Pre-dry: 120°C, minimum 8 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality
- Post-dry: 80–100°C, 1–3 h to increase part strength

Disclaimer

Test Specimens

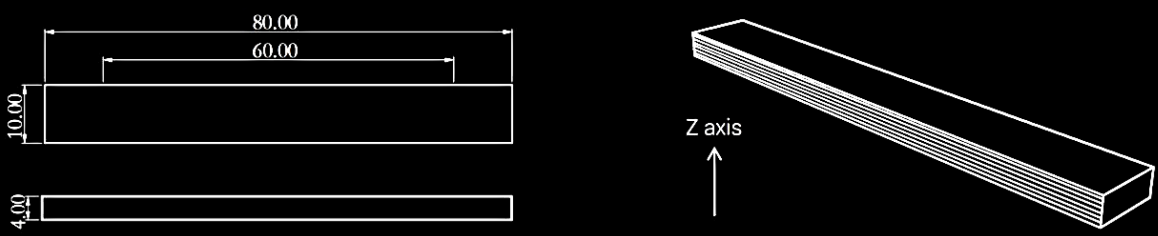
Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

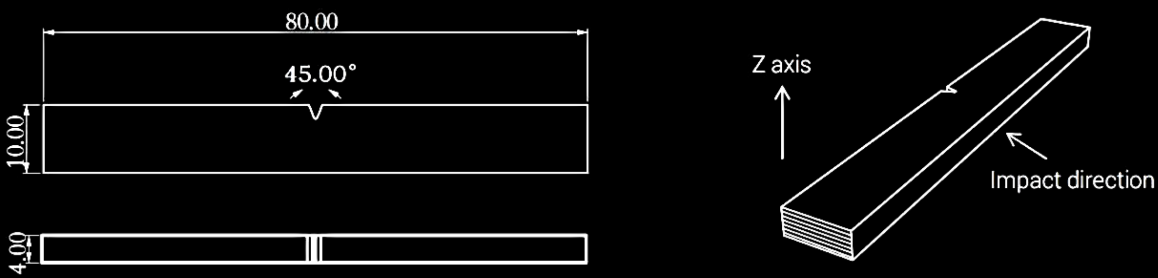


Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)



Flexural testing specimen; ASTMD790 (ISO 178, GB/T 9341)



Impact testing specimen; ASTM D256 (ISO 179, GB/T 1043)